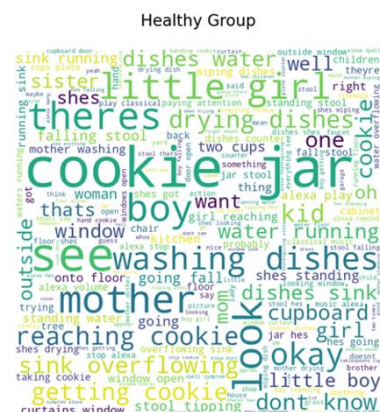
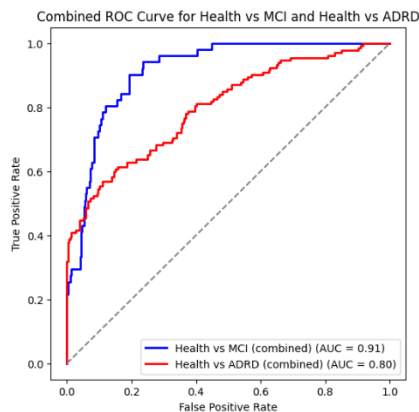
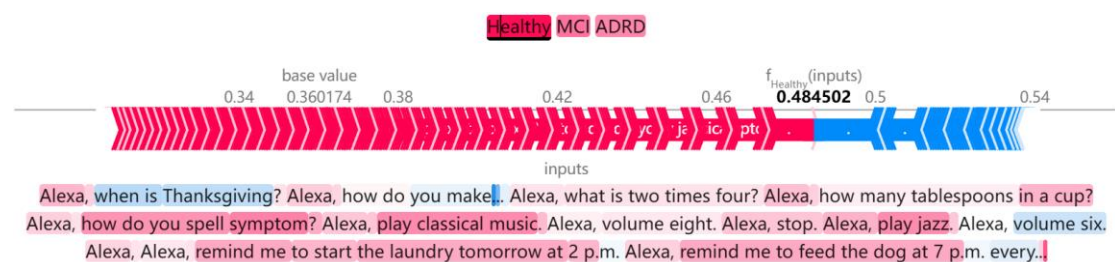


The integrated prediction softmax probability of the multimodal for this example is (0.485, 0.276, 0.239), and the actual label is MCI (Class 2), which is a prediction error.

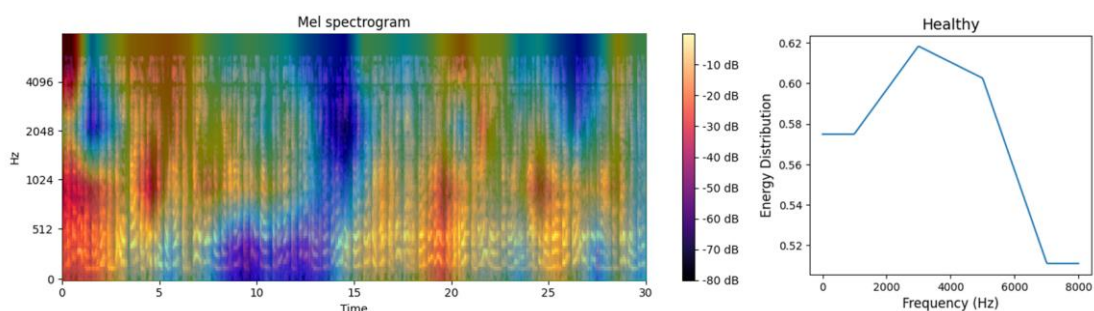
First, according to the ROC curve analysis, the integrated model predicts MCI and ADRD with recall rates of 0.21 and 0.13 respectively when using 0.276 and 0.239 as the threshold. The comparison result shows that the positive rate of MCI is higher. However, the probability of both being predicted as positive is low, and the prediction result is not very credible.



Second, according to the SHAP analysis of the BERT model, the words with medium and high contribution to the model do not appear at all in the word cloud of the healthy category, so they are in the under-trained area of BERT, and the prediction results are not credible.



Finally, according to the CAM analysis of the spectrum graph classification model, it is observed that the model's attention area has high density at low and high frequencies, and is sparse at medium frequencies. The distribution characteristics are completely opposite to the frequency energy statistics of the healthy category, and the prediction results are not credible.



This sample is considered a prediction error and exhibits a relatively consistent lack of credibility in terms of multimodal interpretability. In clinical practice, this could prompt the user to reject the model's prediction at the appropriate time to avoid being misled.